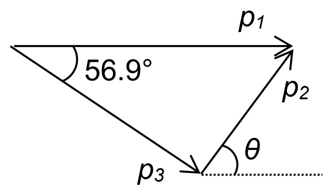
3 **(a)** **(i)**



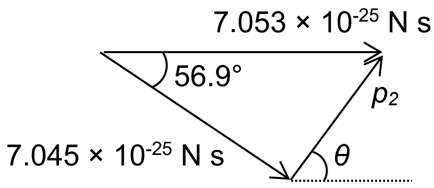
Correct triangle with label and directions [1]
At least 1 angle indicated if correct triangle [1]

(ii) $p_3 = h / \lambda = (6.63 \times 10^{-34}) / (941.1 \times 10^{-12}) = 7.04 \times 10^{-25}$ [1]

(iii) Energy of photon = hc / λ [1]

KE of electron = $hc / (940.0 \times 10^{-12}) - hc / (941.1 \times 10^{-12}) = 2.47 \times 10^{-19}$ [1]

(iv)



Calculating p_1 correctly (i.e. $p_1 = 7.053 \times 10^{-25}$ N s) [1]
Using cosine rule, $p_2 = 6.72 \times 10^{-25}$ N s [1]
Using sine rule, $\sin\theta / (7.045 \times 10^{-25}) = \sin 56.9 / (6.71 \times 10^{-25})$
$\theta = 61.5°$ [1]

4 (a) The product of the force and the perpendicular distance of its line of action from the